



Water quality and eutrophication status of the Zarivar Wetland (Iran)

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Abstract

Wetlands are critically impacted by a variety of pollutants, primarily due to their position at the terminus of both surface and subsurface water flows. This study aims to assess the eutrophication status of Zarivar Wetland, situated in Kurdistan Province, Iran. To evaluate the eutrophication levels, the research employed the Iranian Water Quality Index for Surface Water Resources-Conventional Parameters (IRWQI_{SC}) and the Carlson's Trophic State Index (TSI) during two sampling periods in July and September 2021. The findings indicate that in July, the wetland exhibited an upper-mesotrophic condition, with an average Carlson index value of 52.2, while in September, it transitioned to both upper-mesotrophic and eutrophic conditions, reflected by an average Carlson index value of 60.17. The zoned map of the eutrophication index identified domestic sewage from Marivan city as a significant contributor to elevated eutrophication levels in the southeastern region. Additionally, runoff from irrigated agricultural lands, orchards, and domestic wastewater from surrounding villages contributed to heightened eutrophication in the northeastern and eastern areas of the wetland. A comparative analysis of water quality between July and September 2021 revealed IRWQI_{SC} values ranging from 3.57 to 9.72 in July and from 2.63 to 4.72 in September, categorizing them as relatively good and good, respectively. To safeguard Zarivar Wetland, it is recommended to implement measures that prevent livestock waste discharge into the wetland, manage urban and rural sewage systems, control agricultural runoff, and optimize fertilizer application practices.

Keywords Zarivar Wetland · Agricultural runoff · IRWQI_{SC} · Trophic state index (TSI) · Iran

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Introduction

Worldwide, the water quality of aquatic ecosystems is declining, and ecological functions are degrading (Ding et al. 2020). Increasing water demand, population growth, and environmental degradation have drawn attention to wetlands restoration (Krinsky et al. 2021). The wetland water quality is affected by many factors, including point source pollution (i.e., domestic sewage and industrial discharges, etc.) and non-point source pollution (i.e., agricultural land runoff, atmospheric deposition of pollutants, etc.) (Qiu et al. 2018; Zohrabiet al. 2024). The area of wetlands in the world is decreasing due to drought, socioeconomic development, and land-use changes in the areas around wetlands (Hu et al. 2017). Where wetlands are not directly destroyed by land-use change, external factors like diffuse pollution can reduce wetland water quality that, in turn, can disrupt ecosystem services and functions (Zeng et al. 2015; Alghanimi et al. 2024). A scientific method to assess water quality and to identify potential risk sources is necessary to achieve ecological and environmental management (Huo et al. 2018; Ortiz-Reyes and Anex 2018; Shahradnia et al. 2021). The Iran water quality index (IRWQI) is an emerging national index in Iran to evaluate the quality of water resources. The IRWQI is determined based on the parameters of fecal coliform, biological oxygen demand (BOD_5), chemical oxygen demand (COD), turbidity, total hardness, electrical conductivity (EC), dissolved oxygen, nitrate, ammonium, phosphate, and pH (Shahsavari et al. 2021). This index has been used as case studies in different regions of Iran (Baghapour et al. 2017; Hamed et al. 2015; Nayyeri and Zandi 2015; Sadeghi et al. 2015; Baghapour et al. 2017; Shahsavari et al. 2021).

Eutrophication poses a significant threat to aquatic ecosystems and has garnered global attention (Du et al. 2019). Consequently, classifying wetland ecosystems based on their eutrophication status is crucial. The preservation of habitats for living organisms is essential for effective resource management, particularly through the application of advanced information technologies (Naumoski and Mitreski 2010). The causes of eutrophication are multifaceted, involving a range of ecological, social, and economic factors (Alvarez et al. 2017). Notably, sources such as sewage, agricultural runoff, animal waste, and fertilizers are recognized as primary contributors to increased eutrophication in surface waters (Jani et al. 2020; Lusk et al. 2020). For decades, excessive nutrient presence in aquatic ecosystems has been viewed as a global challenge due to its consequences related to eutrophication (Huang et al. 2020). Over-fertilization can result in nutrient leaching, particularly phosphorus and nitrogen, from agricultural lands into waterways (Kiani et al. 2021). Diffuse pollution

from nitrogen and phosphorus remains the leading cause of eutrophication (Beusen et al. 2016). The rising nutrient concentrations in aquatic ecosystems driven by socioeconomic development and intensified human activities exacerbate eutrophication, which is a critical global environmental issue that threatens the functional integrity of aquatic ecosystems (Wu et al. 2021; Lin et al. 2021; Fielding et al. 2020). Since 1945, indices and models based on determining the eutrophication state of aquatic ecosystems have been designed by many researchers in the world. Several single-parameter and multiparameter indices have been developed for the eutrophication classification of lakes and wetlands (Shannon and Brezonik 1972; Cunha et al. 2013). The most well-known index of them is the trophic state index (TSI), which was proposed for temperate lakes by Carlson (1977) by considering the empirical relations among chlorophyll A, total phosphorus, and Secchi disk depth (Cunha et al. 2013). This index has often been used by researchers and government institutions to indirectly estimate algal biomasses by indicating the degree of eutrophication of aquatic systems (Cunha et al. 2013). The trophic state index (TSI) provides significant management value and essential guidance for lakes with limited data or data constraints (Lin et al. 2020; Bomfim et al. 2019; Stamou et al. 2019). The TSI is determined by analyzing nutrient levels and Secchi disk depth, which serve as indicators of a lake's nutrient status and biological processes (Zhang et al. 2018).

The eutrophication state has been evaluated in the Three-Gorges Reservoir (China) (Xu et al. 2010) using the Carlson Index. It showed that after the reservoir became operational, its condition was oligotrophic ($TSI(S) < 40$) in most months, although both the $TSI(TP)$ and $TSI(TN)$ values were higher than the critical eutrophic state value ($TSI(S) > 50$). In Korea, Mamun and An (2017) worked with 182 agricultural reservoirs and showed the severe phosphorus limitation of these reservoirs. Phosphorus is a crucial nutrient that regulates phytoplankton growth. This relationship was confirmed through an analysis of the indices, demonstrating that $TSI(CHL)$ — $TSI(SD)$ is less than $TSI(CHL)$ — $TSI(TP)$. Dar et al. (2021) in the Brari Nambal urban wetland in Srinagar showed that decreased wetland area, along with a drop in morphometric characteristics, shallow depth, and higher trophic status, indicate anthropogenic pressures using the Carlson TSI (value between 74 and 87) that threaten this wetland's ecological integrity. There are other indices, such as the Lamparelli TSI, used to characterize the eutrophication status of wetlands. Sarkar et al. (2020) analyzed the eutrophication status of the floodplain wetlands of eastern India using the Carlson TSI and Lamparelli TSI. They revealed that the TSI scores had a similar pattern of changes with rainfall and water temperature for both methods. In summary, the

Carlson Index remains one of the best methods for analyzing the eutrophication status of wetlands today.

In Iran, the Zarivar Wetland is one of the largest sources of fresh water. This wetland lies very close to the Marivan city and has been exposed to destruction from natural and human factors. The relevant environmental features and economic and social values of this wetland are noticeable from different perspectives, especially in water pollution terms. Very few studies have focused on assessing the water quality of this wetland to evaluate its suitability for various uses and to identify potential issues. For example, the findings of Ostad-Ali-Askari (2022) showed that the Zarivar international Wetland is heavily polluted with heavy metals that pose significant ecological risks to its environment and human health. In the study of Elahi et al. (2022), the ecological risk of toxic heavy metals was evaluated in the surface sediments of the Zarivar Wetland. The result of risk index values classified the wetland as "very highly polluted" during the rainy season and "considerably polluted" during the dry season. The study of Jamshidi and Naderi (2023) showed that agricultural activities, such as irrigation and land-use change upstream, are the main sources of nitrogen and phosphorus pollution. They suggested that a 50% reduction in fertilizers, animal manures, and irrigation water over a decade can potentially reduce 85% of the nitrogen and 25% of the phosphorus pollution loads per ton of agricultural product. The objectives of the present study were to: (1) assess the extent of the eutrophication status of the Zarivar Wetland; (2) analyze the water quality of the Zarivar Wetland; and (3) evaluate the main factors that cause high eutrophication.

Materials and methods

Study area and sampling

The Zarivar Wetland have been designated as a Wetland of International Importance under the Ramsar Convention. The livelihood of many residents on the edge of the wetland also depends directly or indirectly from it. This wetland is located in the Kurdistan province of Iran, and at between $46^{\circ} 06' 11''$ and $46^{\circ} 09' 16''$ east longitude and between $35^{\circ} 30' 53''$ and $35^{\circ} 35' 12''$ north latitude (north of the folded belt of Zagros). The average height of the Zarivar Wetland is 1285 m above sea level. This wetland was formed due to the severe erosion of the geological formations in this region (Khoram and Hoshmand 2012). In climate terms, the average minimum and maximum temperatures in this wetland are 5 and 20.1 °C, respectively. The average annual rainfall is approximately 900 mm (Anonymous 2016). Figure 1 shows the land-use types in the different parts of the Zarivar Wetland watershed in 2020. A map of land uses and land covers

in the catchment area of the Zarivar Wetland was prepared using images with high spatial resolution of Google Earth Pro and visual interpretation method.

The selection of water sampling stations was carried out systematically to ensure coverage of the entire wetland area. It is worth noting that the peripheral areas of the wetland were not accessible for sampling due to dense vegetation and high concentration of aquatic plants. Sample water was collected from 13 stations in July and September 2021. Samples were transferred to the laboratory at 4 °C on ice and were left in the refrigerator until the start of the analysis. Sterilization of sampling containers, their transportation to, and storage in the laboratory were done according to the instructions in the Standard Method Book, 2017 edition.

Analytical methods and indices

Temperature, EC, pH, and DO parameters were measured using an HQ30d Portable device, model HACH. Turbidity was measured by a turbidity meter (aqualytic model) at the sampling site. Other parameters were transferred to the laboratory for measurements. In the laboratory, the amounts of nitrate (4500-NO₃-E, Cadmium Reduction Method), total phosphate (4500 P D, Stannous Chloride Method), total nitrogen (Persulfate Digestion Method), BOD (BODTrak II™ HACH), COD (COD 5220 C, Closed Reflux, Titrimetric Method), chlorophyll A (10200 H, Spectrophotometric Determination of Chlorophyll), ammonium (4500-NH₃ B&C), and total Coliform (Standard Total Coliform Fermentation Technique 9221 B) were estimated. The quality assurance (QA) of the physicochemical parameters were checked by the spiking method (Adding a known quantity of a specific substance to a sample to assess the performance of an analytical method). The average recoveries for parameters ranged from 94 to 105%. Analytical process precision was evaluated by relative standard deviation (RSD). To fulfill this aim, one water sample was analyzed in three replicates. The RSD values that were obtained for parameters ranged from 0.41% to 6%.

In addition, 86 questionnaires were completed by interviewing the farmers around the wetland with questions about individual social characteristics, economic characteristics, cultivation conditions of agricultural and horticultural crops, and animal husbandry. The estimated amount of fertilizer used in the basin has been fundamentally based on: the analysis done from questionnaires and interviews with farmers; the amount of fertilizer used in the agricultural sector of the wetland research area; and the area where crops are grown.

The IRWQISC

The quality index of Iran's water resources was used to evaluate the quality status of water resources in Iran. The

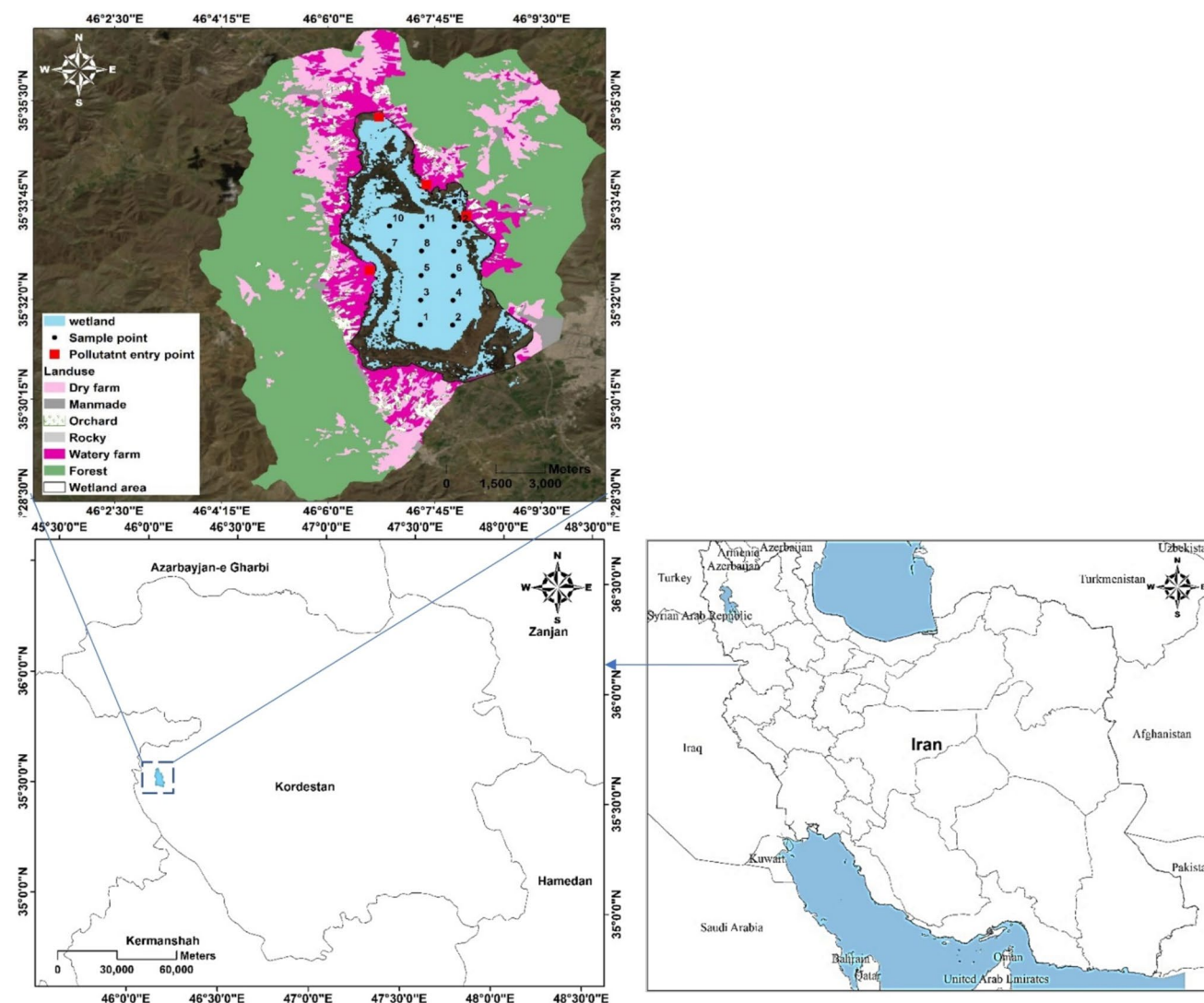


Fig. 1 Location of the Zarivar Wetland in the Kurdistan province and Iran in 2020

Table 1 IRWQI_{SC} parameters and their weights (Hashemi 2011)

Parameter	Weight	Unit
Fecal coliform	0.140	MPN/100 ml
BOD ₅	0.117	mg/l
COD	0.093	mg/l
Nitrate	0.108	mg/l
Phosphate	0.087	mg/l
Ammonium	0.090	mg/l
Dissolved oxygen	0.097	%
EC	0.096	μS/cm
Total hardness	0.059	mg/l (CaCO ₃)
Turbidity	0.062	NTU
pH	0.051	Standard unit

Table 2 Classification of water resources based on the IRWQI_{SC} numerical value (Hashemi 2011)

Index value	Descriptive equivalent
< 15	Very bad
15–29.9	Bad
30–44.9	Relatively bad
45–55	Average
55.1–70	Relatively good
70.1–85	Good
> 85	Very good

water parameters to calculate this index and their weight are presented in Table 1. The description of water quality based on the value of the number obtained from calculating the index is shown in Table 2. Higher values of this index

indicate higher water quality and its calculation are done using the following relations (Hashemi 2011).

$$IRWQI_{sc} = \left[\prod_{i=1}^n I_i^{W_i} \right]^{\frac{1}{\gamma}} \quad (1)$$

$$\gamma = \sum_{i=1}^n W_i \quad (2)$$

where W_i is the weight of the i -th parameter, n is the number of parameters, and I_i is the index value for the i -th parameter of the ranking curve.

Trophic State Index

The Carlson's trophic state index was used to evaluate the eutrophication level of the Zarivar Wetland, of which one of the most reliable is the TSI. This index was presented by Carlson (1977) and modified (Lamparelli 2004). The total amount of TSI is calculated based on the following equations:

$$TSI (SD) = 60 - 14.42 \ln (SD) \quad (3)$$

$$TSI (CHL) = 30.6 + 9.81 \ln (CHL) \quad (4)$$

$$TSI(TP) = 4.15 + 14.42 \ln (TP) \quad (5)$$

$$\text{Total (TSI)} = (TSI (SD) + TSI (CHL) + TSI (TP))/3 \quad (6)$$

where CHL and TP are the concentration of chlorophyll A and total phosphorus in $\mu\text{g/L}$, respectively, and SD is the Secchi disk transparency in meters. The TSI value varies between 1 and 100 (Xu et al. 2010).

Based on the Carlson's trophic state index, the status of wetlands is classified as oligotrophic, mesotrophic, eutrophic, and hypereutrophic (Sarkar et al. 2019). There are other classifications based on this value that classify a wetland more precisely into six classes (Table 3). Based on the total nitrogen to total phosphorus (TN/TP) ratio, the average TSI value (AVG) index is calculated from one of the following formulas:

- If the TN/TP ratio is higher than 30:1, there is phosphorus limitation and the following equation is used:

$$TSI (AVG) = 1/3 [TSI (CHL) + TSI (SD) + TSI (TP)] \quad (7)$$

- If the TN/TP ratio is less than 10:1, there is nitrogen limitation and the following equation is used:

$$TSI (AVG) = 1/3[TSI (CHL) + TSI (SD) + TSI (TN)] \quad (8)$$

$$TSI (TN) = 54.45 + (14.43 \ln (TN)) \quad (9)$$

- If the TN/TP ratio is between 10:1 and 30:1, the environment is in equilibrium condition and the following equation is used:

$$TSI (AVG) = 1/3[TSI (CHL) + TSI (SD) + 0.5(TSI (TP) + TSI (TN))] \quad (10)$$

Statistical and spatial analysis

Descriptive statistics (mean, median, standard deviation, range, etc.) were determined by water parameters and the eutrophication index in July and September of 2021. Although these analysis statistics are essential for analyzing the water quality and the extent of the eutrophication status of the Zarivar Wetland, classic statistical approaches ignore spatial distribution, which contains important information to evaluate the main factors that cause high eutrophication.

Inverse distance weighted (IDW) is one of the most optimal tools to provide an estimate at a non-sampled point. The basic principle of IDW interpolation is to use a weighted linear combination of sample points, which is calculated to create levels and to predict unmeasured points by two statistical and mathematical methods (Khouni et al. 2021). The general equation used for IDW is as follows:

$$\hat{Z}(X_0) = \sum_{i=1}^n Z(X_i) d_{ij}^{-r} / \sum_{i=1}^n d_{ij}^{-r} \quad (11)$$

In the above equation; " $\hat{Z}(X_0)$ " is the estimated value of variable z , " $Z(X_i)$ " is the measured sample value at point X_i , " d_{ij} " is the distance between points i and j , and " r " is an adjustment factor to obtain weight based on distance (Burrough and McDonnell 1998). Mappings were carried out using the geostatistical analyst extension of ArcGIS 10 (ESRI Co., Ltd., Redlands, CA, USA).

Table 3 Classification of eutrophication levels based on the Carlson index (Carlson 1977)

TSI	Eutrophication level
< 30	Oligotrophic
30–40	Lower-mesotrophic
40–50	Mesotrophic
50–60	Upper-mesotrophic
60–70	Eutrophic
70–100	Hypereutrophic

Results and discussion

Water quality of the Zarivar Wetland based on the IRWQISC

Forest cover 49.1%, wetlands 23.9%, farm lands 22.6%, orchards 2.67%, man-made lands 1.67%, and bare lands cover 0.06% of the total area of this Zarivar Wetland watershed.

The summary statistics of water properties and parameters are listed in Table 1. Biological oxygen demand (BOD) ranged from 0.7 to 1.7, with a mean of 1.02 in July and one of 0.17 in September. This parameter is used to determine the pollution potential of the environment and the absorption capacity of the receiving environment based on living organisms' ability to consume dissolved oxygen when they are released to the receiving environment (Anonymous 2013). BOD is defined as the amount of oxygen required for microorganisms to dissolve organic matter in an aqueous environment under aerobic conditions. Another relevant parameter to determine the level of both water and wastewater pollution is chemical oxygen demand (COD) (Mutlu et al. 2013b). The comparison of the mean value of this parameter showed that COD had almost doubled in September (14 mg/l) compared to July (7.13 mg/l). The high COD concentration in summer months may result from the entry of more pollutants from agricultural lands. It may also be due to the presence of many microorganisms in the river (Mirsaeedghazi 2018; Wu et al. 2019). The rapid rise in the COD concentration in summer months is caused by the accumulation of many factors, such as high temperature, low DO concentration, intensive agricultural production, eutrophication, and the presence of wastewater. It is very difficult to identify the main source of pollutions (Kumar et al. 2019; Uddin et al. 2017; Mishra et al. 2017). However, a concentration of more than 25 mg/l COD in waters indicates pollution, while values above 50 mg/l denote severe pollution and possible toxicity to aquatic organisms (Güler 1997). The COD values in the Zarivar Wetland were less than 25 mg/l in both the aforementioned months.

The presence of fecal coliform in aquatic environments may indicate that water has been polluted. The World Health Organization sets a limit of less than 1000 CFU/100 ml of fecal coliforms, which is considered safe for wastewater. In the Zarivar Wetland, the amount of fecal coliform was significantly smaller and decreased in September (87.29 MPN/100 ml) compared to July (518 MPN/100 ml) (Table 4). Ali and Qamar (2013) documented elevated coliform counts during the summer months in the Rawal watershed, attributing this increase to intensified runoff and human activities. However, the

Table 4 The mean measured values of the required parameters of the IRWQISC in the Zarivar Wetland (July and September 2021)

Parameter	July		September	
	Mean	Value range	Mean	Value range
BOD ₅	1.02	0.7–1.7	0.17	0–1
COD	7.13	3–12	14	8–18
Dissolved oxygen	128.01	110–137.2	120.62	109–127.6
Electrical conductivity	440.08	384–482	437.85	414–537
Fecal coliform	518	11–1600	87.29	12–170
Ammonium	0.14	0.032–0.41	0.23	0.12–0.38
Nitrate	3.99	3.39–4.63	3.86	3.66–4.05
Phosphate	0.08	0.06–0.11	0.098	0.03–0.16
Total hardness	183.62	172–194	190.23	186–194
Turbidity	4.04	2.4–11.3	5.15	4.16–6.6
pH	7.38	6.94–7.7	7.63	7.33–8

results of the current study reveal fluctuations in contamination severity, influenced by the sampling periods and potentially local environmental conditions. Temperature and solar radiation are considered the most important abiotic factors (Kagalou et al. 2002; Xu et al. 2002) that affect the mortality rate caused by fecal coliform. Water temperature changes are caused by seasonal temperature changes (Mutlu et al. 2013a). Burkhardt et al. (2000) found in their study that sunlight or temperature accounted for 99% of the reduction in fecal coliform levels in estuarine waters of Alabama and on Rhode Island. In the present study, as the mean water temperature in the sampled stations was 28.4 °C in July and 26.7 °C in September, one of the reasons for the smaller amount of fecal coliforms in September compared to July can be attributed to the drop in water temperature.

Dissolved oxygen is an essential compound for aquatic life and is also necessary for biochemical oxidation. In freshwater, there must be at least 5 mg/l of dissolved oxygen to sustain aquatic life (Atay and Pulatsü, 2000). The value of this parameter was between 9.3 and 18.7 mg/l in July and between 7.8 and 7.23 mg/l in September (Table 4). These values show that there was minimal dissolved oxygen in all the samples. EC is also very important for aquatic products. When conductivity passes the level of 100 µS/cm, pollution capacity increases (Verep et al. 2005) which, according to the results of the EC values in this study, it was more than 100 µS/cm in both months. Both temperature and water salinity increased after taking measurements in summer.

According to the United States Environmental Protection Agency (USEPA 1986), the suitable pH range for wetland water is 6.5–9. This pH range supports the growth and survival of a diverse array of aquatic plants, fish, and other organisms that thrive in wetland environments. EPA guidelines state that pH levels beyond this range can negatively

impact the balance of wetland ecosystems, which can potentially lead to issues like biodiversity loss, algal blooms, and poorer water quality. According to the results, the range of pH values of the water samples in both months fell within the standard range. In general, changes in water quality parameters can significantly impact the $IRWQI_{SC}$ and water quality. For example, high BOD and COD levels indicate a high degree of organic pollution, which can negatively impact water quality (Fallah et al. 2021). Similarly, higher EC values indicate increased levels of dissolved solids (Fallah et al. 2021), while rising total hardness values depict higher levels of soluble minerals (Umar et al. 2021). These changes can lead to worse water quality and low $IRWQI_{SC}$ value. Furthermore, low dissolved oxygen levels can diminish water quality (Fallah et al. 2021). The presence of fecal bacteria is indicated by high fecal coliform levels, while increased levels of ammonium suggest rising levels of organic matter and nutrient pollution (Bravo de Guenni et al. 2024). High nitrate and phosphate levels also indicate marked nutrient pollution, which is dangerous for aquatic life and degrades water quality (Bravo de Guenni et al. 2024). Additionally, higher turbidity levels can reduce water clarity and quality by increasing the amount of suspended solids (Umar et al. 2021).

Based on the IDW maps (Fig. 2), high $IRWQI_{SC}$ values (better water quality) were observed in the northwestern, southeastern, southwestern, and then the northeastern, parts of the wetland in July, and in the southwestern and northwestern parts in September. The $IRWQI_{SC}$ in these 2 months showed that the of $IRWQI_{SC}$ values range in July was between 57.3 and 72.9 and in the relatively good condition, and between 63.2 and 72.4 and in the good condition in September. Water quality was lower in the northeastern part of the wetland in July and in the western part of the wetland in September. The water quality parameters from the present study indicate that the low water quality in the northeastern part of the wetland in July was primarily due to elevated levels of nitrate and fecal coliform. In September, the decline in water quality was attributed to high nitrate levels and low dissolved oxygen compared to other seasons. The results of the present study are in agreement with the findings of Ebrahimpour et al. (2011), in their study reported that most of the polluting sources in the north and east are non-point sources, such as agricultural activities and domestic sewage. Given the fact that a large amount of sediment enters the wetland from the north every year, there is more suspended matter in that area that causes the visibility depth of the Sashi disk to decrease. Polluting substances are also brought in from all around the wetland by seasonal waterways, which are effective in accelerating eutrophication and advancing reeds. The Zarivar Wetland had worse water quality in the western part in July due to the high levels of ammonium, BOD_5 and fecal coliform, and the low dissolved oxygen level compared

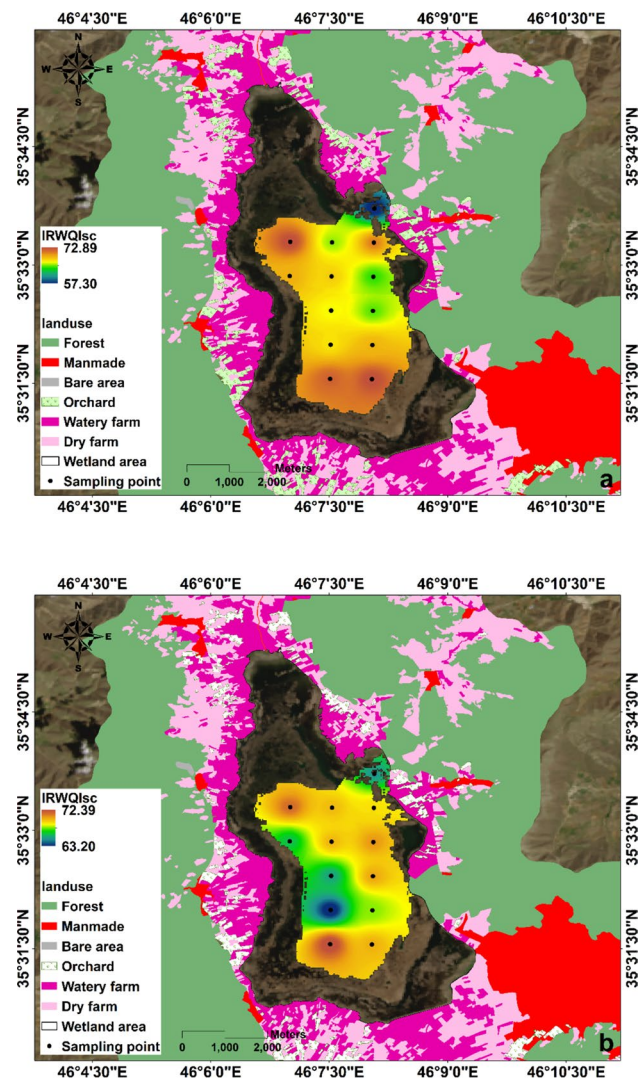


Fig. 2 Zoned map of the $IRWQI_{SC}$ in the Zarivar Wetland in **a** July and **b** September 2021

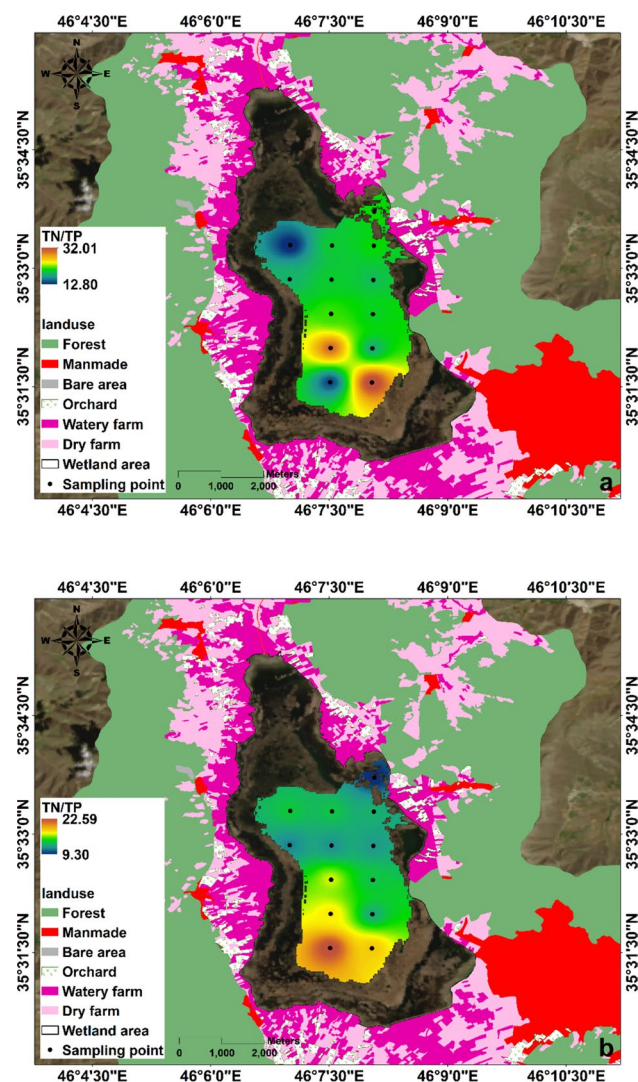
to other seasons. In September, this was due to high EC, total hardness, fecal coliform and phosphate compared to other seasons, and the western part had low water quality. When generally considering the land use around the wetland, the reasons for the lower water quality in these parts could be attributed to sewage from the residential areas of the villages without a sewage network and to the use of chemical fertilizers on agricultural lands.

Extent of the eutrophication status of the Zarivar Wetland

Table 5 shows the main parameters associated with the Zarivar Wetland eutrophication. The first thing that was observed is that the amount of chlorophyll and total phosphorus significantly increased in September compared

Table 5 The mean values of the measured parameters of the eutrophication index in the Zarivar Wetland (July and September 2021)

Parameter	July		September	
	Mean	Parameter value range	Mean	Parameter value range
SD (m)	1.53	1.15–2	1.24	1–1.5
Chl-a ($\mu\text{g/l}$)	1.78	0.33–4.09	4.38	1.59–8.05
TP ($\mu\text{g/l}$)	211.15	128–331	277.58	100.5–462
TN (mg/l)	4.13	3.47–4.7	4.06	3.78–4.31

**Fig. 3** Zoned map of the nitrogen to phosphorus ratio in the Zarivar Wetland in **a** July and **b** September 2021

to July. According to the zoned map of the TN/TP ratio (Fig. 3), in July, its maximum was observed in the southeastern part, which is more than 30 and, therefore, has phosphorus limitation. The southeastern part of the wetland is affected mostly by the domestic sewage of the Marivan city, which includes the largest residential area around the Zarivar Wetland, and high N/P ratios due to

nitrogen loading can reduce phosphorus availability in relation to nitrogen, which can potentially lower net phosphorus mineralization rates from decomposition (Luo et al. 2022). The TN/TP ratio was less than 10 in the northeastern part of the wetland in September and, therefore, has nitrogen limitation. This part of wetland is also affected mostly by irrigated agricultural lands and gardens, and domestic sewage from the villages of Kolan, Mohammadeh and Seyf-e Sofla, which have no sewage network. The maximum phosphorus value was also observed in this part of the wetland. There is also a waterway to the east of the basin that directly enters the wetland. Residues of fertilizers and chemical toxins from agricultural lands in this part of the wetland are washed away by rain and can enter the wetland by the waterway. In the study of Jamshidi and Naderi (2023), mainly agricultural activities like irrigation and land-use changes upstream were mentioned as being responsible for TN and TP pollution loads. During other seasons, this ratio was between 10 and 30, which is marked by green, and they were in equilibrium with no nitrogen and phosphorus limitations. It should be noted in July that a part of the western wetland is close to the phosphorus limitation, and a northwestern part comes close to the nitrogen limitation.

The spatial TSI (AVG) eutrophication index distribution in the Zarivar Wetland (Fig. 4) showed a good spatial correlation with the TSI indices that are found in Table 6. The wetland's eutrophication indicates a stronger influence in the northeastern and southeastern parts in July and in the east, west, and northwestern parts of the wetland in September. The eutrophication in the eastern half of the wetland was generally higher than its western half in July. This spatial pattern suggests the presence of several local causes of variability due to point sources of pollution from three villages without a sewage network (Kolan, Mohammadeh, and Seyf-e Sofla) in the east part of the study area, and the wetland is affected by effluents from these villages. In the east, west, northeastern, and then the southeastern parts of the wetland, eutrophication increased in September compared to July, and these areas were in a eutrophic state. This provides the slope of the agricultural lands with spatial refinement, and phosphate and nitrate fertilizers are erased from the surface of these lands after rainfall and enter the wetland with flood flows.

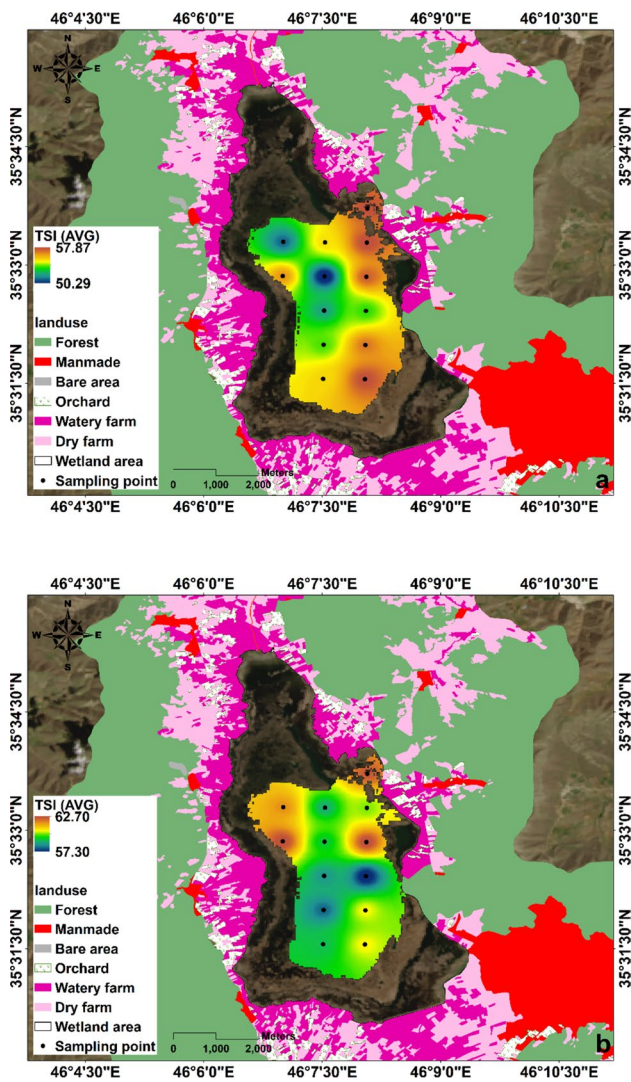


Fig. 4 Zoned map of eutrophication index in the Zarivar Wetland in **a** July and **b** September 2021

Table 6 TSI indices in the Zarivar Wetland in July and September 2021

Index	July		September	
	Mean	Value range	Mean	Value range
TSI (SD)	54	50.02–57.99	56.98	54.16–60
TSI (CHL)	33.65	19.25–43.81	43.7	34.52–50.46
TSI (TP)	80.82	74.07–87.77	84.46	70.58–92.57
TSI (TN)	56.15	51.33–58.83	74.66	73.65–75.53
TSI (Total)	74.86	72.4–76.77	61.71	58.7–65.22
TN/TP	20.75	12.8–32	14.41	9.33–22.61
TSI (AVG)	55.15	50.28–57.87	60.17	57.26–62.7

The eutrophication state of the Zarivar Wetland based on the TSI (AVG) and TSI (total) results indices is shown in

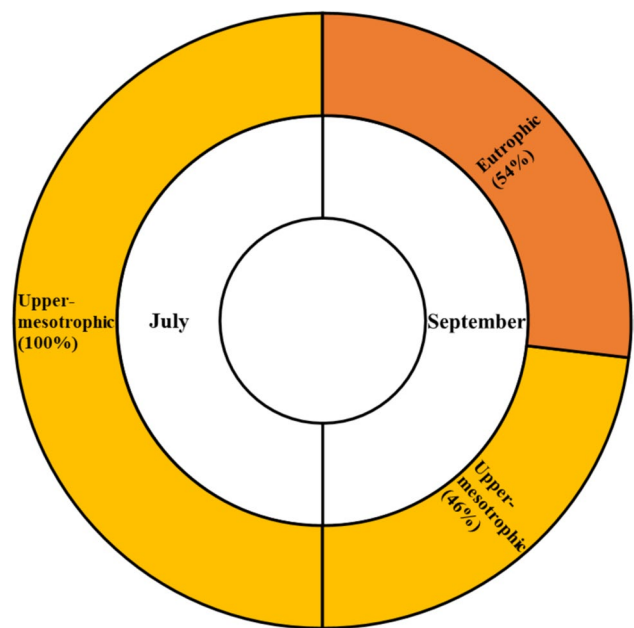


Fig. 5 Diagram of the eutrophication status of the Zarivar Wetland in July and September 2021

Fig. 5. The values of these indices were between 50 and 60 in all the stations in July 2021, which means that the wetland was in an upper-mesotrophic state. In September, 46% of the wetland was in an upper-mesotrophic state and 54% was in an eutrophic state.

Based on the results of the present study, there are three primary pollution groups related to the eutrophication status of Zarivar Wetland. This pattern corresponds to livestock, agricultural practices, and villages without a sewage network. The approximate number of light and heavy livestock in the villages around the wetland is 9770 and 2003, respectively. When the excreta of each livestock (cow) and light livestock (sheep and goat) as 8.99 and 1.16 kg per day is considered as 8.99 and 1.16 kg per day, respectively (Ministry of Agriculture Jihad of Iran, 2019), the amount of light and heavy livestock excreta is estimated at 4137 and 6573 tons per year, respectively. If considered agricultural and orchard lands in the Zarivar Wetland watershed (2540 hectares) and the average amount of chemical fertilizers used on agricultural lands being about 100 kg per hectare annually (based on the results of interviews held with farmers and the analysis of questionnaires), we could estimate that the chemical fertilizers use in the watershed of the wetland is estimated at 254 tons per year. Furthermore, according to the Iran's Management and Planning Organization, about 80–90% (on average 85%) of the water used for sanitary-human purposes is turned into sewage. Therefore, considering the average per capita water consumption per person per day in rural areas (100 l per day) and taking into account the average conversion factor of water to wastewater (0.85), the

volume of wastewater produced by the villages of Kolan, Seyf-e Sofla, and Mohammadeh, given their populations, is 26,371.25, 15,946.85, and 5646.55 cubic meters per year, respectively. It can be concluded that the volume of sewage produced in the villages of Kolan, Seyf-e Sofla, and Mohammadeh is totally 47,964.65 cubic meters per year. By considering the populations of the three villages without a sewage network on the eastern edge of the wetland (1546 people), and by bearing in mind the human production of nitrogen and phosphorus at around 10 and 4 g per person per day, respectively, the amount of nitrogen produced in these villages is 16.2 kg. per day, and the amount of phosphorus comes to 6.48 kg per day.

The findings of this study contribute to the expanding body of knowledge regarding wetland eutrophication. Specifically, the trophic state index (TSI) values observed in Zarivar Wetland, which transitioned from mesotrophic to eutrophic conditions, are comparable to those reported by Jarosiewicz et al. (2011) in their assessment of Pomeranian lakes. They identified total phosphorus as a primary determinant of trophic state, underscoring the significance of nutrient management in mitigating eutrophication. This observation is supported by our results, which indicate a strong correlation between total phosphorus levels and elevated TSI values during the study period. Moreover, the seasonal increase in TSI observed in this study—most pronounced in September—mirrors patterns observed in Lake Simbi, as described by Opiyo et al. (2019), where seasonal nutrient inflows from agricultural and urban sources exacerbated eutrophic conditions. These findings highlight the impact of external nutrient loading on wetland ecosystems and reveal common challenges across various geographic contexts. Additionally, the spatial distribution of eutrophication within Zarivar Wetland, primarily driven by agricultural runoff and untreated domestic sewage, aligns with global observations. The studies on Pomeranian lakes similarly identified agricultural practices and urban waste as significant contributors to nutrient enrichment, further corroborating the drivers identified in this research. Overall, this study provides valuable insights into the eutrophication status of Zarivar Wetland through the application of the $IRWQI_{SC}$ and Carlson's Trophic State Index. However, certain limitations warrant consideration. The analysis was confined to two specific sampling periods, which restricts the ability to comprehensively capture seasonal variations. While the indices employed are robust for assessing eutrophication, incorporating additional factors such as soil and wind erosion, sediment deposition, and detailed nutrient fluxes could enhance the accuracy of future evaluations. Caution should be exercised when generalizing these findings to other wetlands due to significant variability in local conditions, pollution sources, and management practices. Future research should focus on year-round sampling, advanced modeling

techniques, and broader spatial coverage to achieve a more comprehensive understanding of eutrophication dynamics and their underlying drivers. Addressing regional challenges related to agricultural drainage and wastewater management is crucial for preserving wetland ecosystems like Zarivar Wetland and preventing further eutrophication. Expanding future research efforts to include a broader temporal scope or cross-ecosystem comparative analyses could further refine management practices aimed at mitigating nutrient loading.

Conclusion

This study evaluated the eutrophication status of Zarivar Wetland utilizing trophic state index (TSI) indices and spatial mapping techniques. The findings indicated that the wetland was in an upper-mesotrophic state in the summer of July 2021, with an average Carlson index of 52.2, and transitioned to both upper-mesotrophic and eutrophic conditions by September (early autumn) 2021, reflected in an average Carlson index of 60.17. Spatial analyses identified significant pollution sources, including agricultural runoff, domestic sewage from adjacent villages, and excessive livestock grazing, which contributed to increased soil erosion and nutrient loading. Elevated total nitrogen levels at sampling stations 12 and 13 were associated with the intensive application of chemical fertilizers in surrounding agricultural areas. To maintain the ecological balance of the wetland, effective management strategies are essential, including controlling agricultural drainage, reducing livestock waste, and regulating fertilizer application. However, this study has certain limitations, such as the absence of data on soil and wind erosion and its impact on nutrient dynamics, as well as a lack of seasonal variability beyond the months studied. Future research should investigate these factors to provide a more comprehensive understanding of eutrophication processes. By addressing these gaps, more effective strategies for the restoration and management of this wetland can be developed.

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Authors contribution Data collection was performed by Rasool Zamani-Ahmadmahooodi, Samira Bayati, Khodayar Abdollahi, Davood Mafi-Gholami, and Arezou Raeisi-bidekani. Rasool Zamani-Ahmadmahooodi, Samira Bayati, Khodayar Abdollahi, Sharif Joorabian Shoostari, and Seyedeh Samira Soleimanipour supervised the research, designed the experiments, analyzed the data, and cowrote the paper.

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Availability of data and materials The datasets applied and/or analyzed throughout the present research are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethics approval and consent to participate Not applicable.

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